

# **EE8203 - Design and Management of Data Networks**

Department-Level Campus Network Design &  
Management

Group Project - Design Document

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**Version:** 1.0

# 1. Department Design

This document presents the network design for the Faculty of Engineering at the University of Ruhuna, Hapugala premises. The design interconnects four academic and administrative departments:

| Code | Department                                      | Role                                            |
|------|-------------------------------------------------|-------------------------------------------------|
| DEIE | Dept. of Electrical & Information Engineering   | Engineering research and undergraduate labs     |
| DCEE | Dept. of Civil & Environmental Engineering      | Administration and project coordination offices |
| DMME | Dept. of Mechanical & Manufacturing Engineering | Workshop lab and server access zone             |
| DIS  | Dept. of Inter Disciplinary Studies             | Faculty IT hub - hosts central server farm      |

The network follows a **three-layer hierarchical design** (Core, Distribution, Access) with VLAN segmentation, OSPF routing, NAT, and strict ACL-based security policies. The design supports automation via Python/Netmiko and Ansible, and monitoring via Zabbix.

## 2. Network Design Overview

### 2.1 Design Philosophy

The network is designed following industry best practices:

- **Hierarchical Design:** Three-layer model (Core, Distribution, Access) for scalability and fault isolation
- **VLAN Segmentation:** Separate VLANs for each department and management
- **Redundancy:** Loopback interfaces for OSPF stability
- **Security:** ACLs enforce inter-department access policies
- **Automation:** Infrastructure supports Netmiko and Ansible for configuration management
- **Monitoring:** Zabbix with SNMP provides visibility into network health

2.2 Network Hierarchy

| Layer        | Devices                                   | Function                                                  |
|--------------|-------------------------------------------|-----------------------------------------------------------|
| Core         | R-CORE, R-EDGE, SW-CORE                   | High-speed backbone, inter-VLAN routing, WAN connectivity |
| Distribution | SW-D-DEIE, SW-D-DCEE, SW-D-DMME           | Aggregation, policy enforcement, departmental gateway     |
| Access       | SW-A-DEIE, SW-A-DCEE, SW-A-DMME, SW-A-DIS | End-user connectivity, port security                      |

3. IP Addressing Plan

3.1 Addressing Scheme Overview

The network uses private IP address space (10.0.0.0/8) with subnetting based on departmental requirements.

3.2 VLAN Subnets

| VLAN ID | VLAN Name | Department  | Subnet        | Gateway    | Purpose                           |
|---------|-----------|-------------|---------------|------------|-----------------------------------|
| 10      | VLAN_DEIE | DEIE        | 10.10.10.0/24 | 10.10.10.1 | Engineering workstations          |
| 20      | VLAN_DCEE | DCEE        | 10.10.20.0/24 | 10.10.20.1 | Civil/Environmental dept hosts    |
| 30      | VLAN_DMME | DMME        | 10.10.30.0/24 | 10.10.30.1 | Mechanical/Manufacturing hosts    |
| 40      | VLAN_DIS  | DIS         | 10.10.40.0/24 | 10.10.40.1 | Faculty IT hub / server zone      |
| 99      | VLAN_MGMT | All devices | 10.99.99.0/24 | 10.99.99.1 | Out-of-band management (SSH only) |
| 100     | NATIVE    | Trunk links | N/A           | N/A        | Native VLAN (untagged frames)     |

3.3 Point-to-Point Links

| Link                | Network          | R-EDGE IP     | R-CORE IP  | SW-CORE IP |
|---------------------|------------------|---------------|------------|------------|
| R-EDGE ↔ ISP-Router | 200.200.200.0/30 | 200.200.200.2 | -          | -          |
| R-EDGE ↔ R-CORE     | 10.255.0.0/30    | 10.255.0.1    | 10.255.0.2 | -          |
| R-CORE ↔ SW-CORE    | 10.255.0.4/30    | -             | 10.255.0.5 | 10.255.0.6 |

### 3.4 Management IP Addresses

| Device    | Management IP | VLAN | Interface   |
|-----------|---------------|------|-------------|
| SW-CORE   | 10.99.99.1    | 99   | Vlan99      |
| SW-D-DEIE | 10.99.99.11   | 99   | Vlan99      |
| SW-A-DEIE | 10.99.99.21   | 99   | Vlan99      |
| SW-D-DCEE | 10.99.99.12   | 99   | Vlan99      |
| SW-A-DCEE | 10.99.99.22   | 99   | Vlan99      |
| SW-D-DMME | 10.99.99.13   | 99   | Vlan99      |
| SW-A-DMME | 10.99.99.23   | 99   | Vlan99      |
| SW-A-DIS  | 10.99.99.24   | 99   | Vlan99      |
| R-EDGE    | 10.255.0.1    | -    | Serial0/1/0 |
| R-CORE    | 10.255.0.5    | -    | Gig0/0/0    |

### 3.5 End-Device IP Assignments

| Device          | VLAN | IP Address  | Subnet Mask   | Gateway    |
|-----------------|------|-------------|---------------|------------|
| DEIE-PC1        | 10   | 10.10.10.10 | 255.255.255.0 | 10.10.10.1 |
| DEIE-PC2        | 10   | 10.10.10.11 | 255.255.255.0 | 10.10.10.1 |
| DCEE-PC1        | 20   | 10.10.20.10 | 255.255.255.0 | 10.10.20.1 |
| DCEE-PC2        | 20   | 10.10.20.11 | 255.255.255.0 | 10.10.20.1 |
| DMME-PC1        | 30   | 10.10.30.10 | 255.255.255.0 | 10.10.30.1 |
| DMME-PC2        | 30   | 10.10.30.11 | 255.255.255.0 | 10.10.30.1 |
| DIS-PC1         | 40   | 10.10.40.10 | 255.255.255.0 | 10.10.40.1 |
| DIS-PC2         | 40   | 10.10.40.11 | 255.255.255.0 | 10.10.40.1 |
| DHCP/WEB Server | 40   | 10.10.40.2  | 255.255.255.0 | 10.10.40.1 |
| DNS Server      | 40   | 10.10.40.3  | 255.255.255.0 | 10.10.40.1 |
| Google Server   | -    | 8.8.8.8     | 255.255.255.0 | 8.8.8.1    |

Later Configured DHCP Server then got dynamic ip addresses for the vPCs

## 4. VLAN Scheme

### 4.1 VLAN Allocation

| VLAN ID | Name      | Type       | Purpose                                                |
|---------|-----------|------------|--------------------------------------------------------|
| 10      | VLAN_DEIE | User       | Department of Electrical & Information Engineering     |
| 20      | VLAN_DCEE | User       | Department of Civil & Environmental Engineering        |
| 30      | VLAN_DMME | User       | Department of Mechanical & Manufacturing Engineering   |
| 40      | VLAN_DIS  | Server     | Department of Inter Disciplinary Studies / Server Farm |
| 99      | VLAN_MGMT | Management | Out-of-band management network                         |
| 100     | NATIVE    | Native     | Untagged frames on trunk links                         |

### 4.2 VLAN Trunking

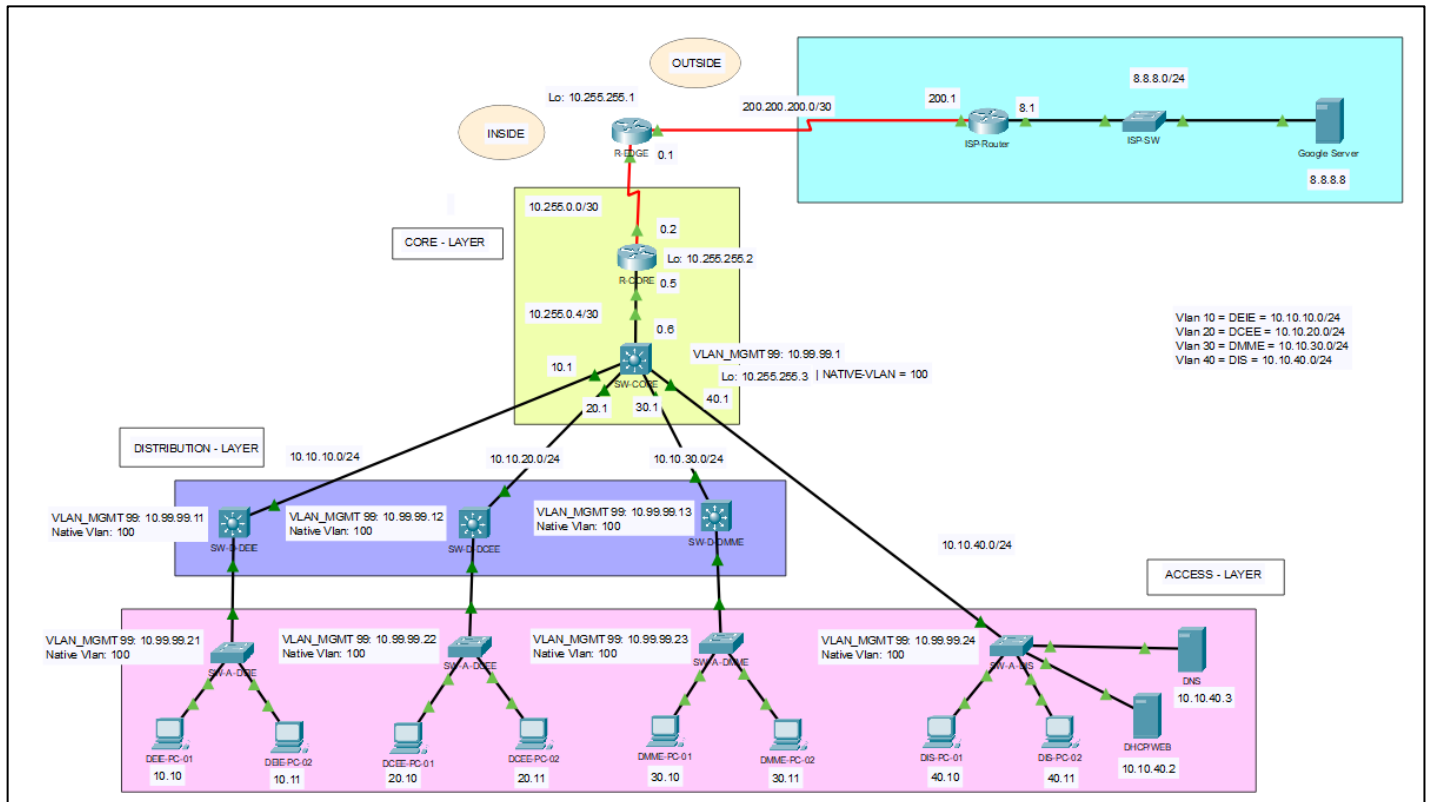
All inter-switch links are configured as **802.1Q trunks** with (MLSW)

- **Native VLAN:** 100 (untagged frames)
- **Allowed VLANs:** Department VLAN + Management VLAN + Native VLAN
- **Encapsulation:** dot1Q (default on Cisco switches) (Routers and other L2 switches)

#### Trunk Allowed VLANs

| Link                  | Allowed VLANs |
|-----------------------|---------------|
| SW-CORE ↔ SW-D-DEIE   | 10, 99, 100   |
| SW-CORE ↔ SW-D-DCEE   | 20, 99, 100   |
| SW-CORE ↔ SW-D-DMME   | 30, 99, 100   |
| SW-CORE ↔ SW-A-DIS    | 40, 99, 100   |
| SW-D-DEIE ↔ SW-A-DEIE | 10, 99, 100   |
| SW-D-DCEE ↔ SW-A-DCEE | 20, 99, 100   |
| SW-D-DMME ↔ SW-A-DMME | 30, 99, 100   |

## 5. Topology Diagram



**Figure 1: Topology Diagram V-2**

Note: Here we used ISP-Router and ISP-Switch to test the connectivity with the google server in the OUTSIDE/WAN side

## 6. Device Selection Justification

### 6.1 Device Inventory (GNS3)

| Device / Hostname                         | Qty | Image / Type   | Primary Role                                     |
|-------------------------------------------|-----|----------------|--------------------------------------------------|
| R-CORE                                    | 1   | IOSv           | Core router, OSPF area 0, NAT                    |
| R-EDGE                                    | 1   | IOSv           | Edge / WAN router, default route, NAT overload   |
| SW-CORE                                   | 1   | IOSvL2         | Core L3 switch, inter-VLAN routing backbone      |
| SW-D-DEIE, SW-D-DCEE, SW-D-DMME           | 3   | IOSvL2         | Distribution L3 switches                         |
| SW-A-DEIE, SW-A-DCEE, SW-A-DMME, SW-A-DIS | 4   | IOSvL2         | Access L2 switches                               |
| VM-AUTO                                   | 1   | Linux (Ubuntu) | Automation host - Python/Netmiko + Ansible + SSH |
| VM-ZABBIX                                 | 1   | Linux (Ubuntu) | Zabbix server, SNMPv2c monitoring                |
| VM-DHCP/WEB                               | 1   | Linux          | DHCP/DNS server; optional web server             |
| VPC end-hosts                             | 8+  | GNS3 VPCs      | Two per department - used for ACL policy testing |

### 6.2 Device Selection Rationale

#### Core Layer - R-CORE and R-EDGE (IOSv)

- OSPF Support: Full OSPF implementation required for dynamic routing
- NAT Capability: Supports NAT overload (PAT) for internet egress
- Scalability: Handles routing for the entire campus network

#### Why Cisco IOSv?

- IOSv is lightweight and runs efficiently in virtual environments
- Provides full enterprise routing features needed for the design
- Familiar CLI reduces implementation errors

#### Core Switch - SW-CORE (Cisco IOSvL2)

- Layer 3 Switching: Required for inter-VLAN routing at wire speed
- OSPF Routing: Participates in OSPF area 0 for dynamic routing
- VLAN Support: Supports VLAN creation, trunking, and SVIs
- Performance: Handles traffic between all four departments efficiently

## Distribution Switches - SW-D (Cisco IOSvL2)

- Aggregation: Aggregates access layer switches for each department
- VLAN Trunking: Supports 802.1Q trunking
- Management: Each distribution switch has a management IP on VLAN 99
- Scalability: Allows adding more access switches per department

## Access Switches - SW-A (Cisco IOSvL2)

- End-User Connectivity: Provides access ports for end devices
- Port Security: Supports Portfast and BPDUguard for security
- VLAN Assignment: Assigns devices to appropriate departmental VLANs
- Cost-Effective: Simple L2 switches are sufficient for access layer

## Virtual Machines (Ubuntu)

- VM-AUTO: Linux is ideal for automation (Python, Netmiko, Ansible)
- VM-ZABBIX: Zabbix server runs natively on Linux
- VM-DHCP/WEB: Linux provides flexible DHCP and web server options
- Cost-Effective: Open-source solutions reduce licensing costs

## 6.3 Tool Selection Rationale

### Netmiko, Python for Routers

- Device-Specific: Handles Cisco IOS (prompts, paging)
- Flexibility: Python-based for custom logic and error handling
- Integration: Works well with inventory files (JSON/YAML)

### Ansible for Switches

- Built-in functions prevent duplicate configs
- Scalability: Designed for large infrastructure automation
- Roles: Organizes configuration by function (VLANs, trunks, STP)
- Community: Wide adoption and extensive module library

### Why Different Tools for Different Tasks?

| Aspect          | Routers (Netmiko)             | Switches (Ansible)           |
|-----------------|-------------------------------|------------------------------|
| Task Complexity | Complex (NAT/PAT, OSPF, ACLs) | Structured (VLANs, trunking) |
| Scalability     | Good for small-mid scale      | Excellent for large scale    |
| Learning Curve  | Moderate                      | Steeper                      |
| Configuration   | Python scripts                | YAML playbooks               |

# 7. Network Architecture Details

## 7.1 Routing Protocol - OSPF

OSPF (Open Shortest Path First) is used as the dynamic routing protocol:

| Parameter       | Value                                                 |
|-----------------|-------------------------------------------------------|
| OSPF Process ID | 1                                                     |
| Area            | 0 (Single Area)                                       |
| Router IDs      | 1.1.1.1 (R-EDGE), 2.2.2.2 (R-CORE), 3.3.3.3 (SW-CORE) |

**OSPF Network Statements:**

| Device  | Network      | Wildcard Mask | Area |
|---------|--------------|---------------|------|
| R-EDGE  | 10.255.0.0   | 0.0.0.3       | 0    |
| R-EDGE  | 10.255.255.1 | 0.0.0.0       | 0    |
| R-CORE  | 10.255.0.0   | 0.0.0.3       | 0    |
| R-CORE  | 10.255.0.4   | 0.0.0.3       | 0    |
| R-CORE  | 10.255.255.2 | 0.0.0.0       | 0    |
| SW-CORE | 10.255.0.4   | 0.0.0.3       | 0    |
| SW-CORE | 10.10.10.0   | 0.0.0.255     | 0    |
| SW-CORE | 10.10.20.0   | 0.0.0.255     | 0    |
| SW-CORE | 10.10.30.0   | 0.0.0.255     | 0    |
| SW-CORE | 10.10.40.0   | 0.0.0.255     | 0    |
| SW-CORE | 10.99.99.0   | 0.0.0.255     | 0    |
| SW-CORE | 10.255.255.3 | 0.0.0.0       | 0    |

## 7.2 NAT Configuration

| Component         | Configuration                                              |
|-------------------|------------------------------------------------------------|
| Inside Interfaces | Serial0/1/0 (R-EDGE to R-CORE)                             |
| Outside Interface | Serial0/1/1 (R-EDGE to ISP)                                |
| NAT ACL           | Standard ACL 1 permits 10.10.10.0/24 and 10.10.20.0/24     |
| NAT Type          | PAT (Port Address Translation) / Overload                  |
| Translation       | ip nat inside source list 1 interface Serial0/1/1 overload |